**ORIE 4154/5154: Pricing and Market Design** **Fall 2026**

# Lecture 11 — Compensated Coupling and the Bayes Selector

*Lecturer: Sid Banerjee* *Date: September 29; October 6 and 8, 2026*

*Meeting record.* Lecture 11 (September 29, Zoom) introduced single-resource allocation and the hindsight comparison. Lecture 12 (October 1) was canceled. Lecture 13 (October 6) developed the uniform-value compensated-coupling proof. Lecture 14 (October 8) developed the finite-type Bayes Selector proof and introduced substitution. The Spiral-Down Effect is the topic of Lecture 15 (October 15), with separate notes.

## 11.1 Why compare online decisions with hindsight?

Lectures 9–10 gave an $O(\sqrt{T})$ additive guarantee for a static fluid policy. Their two-value example shows a matching gap between that particular policy and hindsight. Realized demand fluctuates around its mean; an online policy can respond to these fluctuations as they unfold. We ask:

| When can an online decision agree with an optimal hindsight decision? |
|---|

The argument here follows the compensated-coupling structure of Banerjee's 2022 Simons Institute talk, specialized to the model taught in class.

## 11.2 One resource and three benchmarks

There are $T$ periods and initial capacity $B \in \{0, \dots, T\}$. Buyer $t$ has random value $V_t \geq 0$; the values are independent and identically distributed (iid) according to known $F$. After seeing $V_t$ and the past, a policy chooses $A_t \in \{0, 1\}$, where 1 means accept. Each acceptance uses one unit. Its realized reward is $R^\pi = \sum_{t=1}^{T} A_t V_t$, with $\sum_t A_t \leq B$.

For realized values $v_{(1)} \geq \cdots \geq v_{(T)}$ in descending order, the offline clairvoyant earns

$$\mathrm{OPT}(v_1, \dots, v_T) = \sum_{i=1}^{B} v_{(i)}.$$

The empty sum is zero when $B = 0$; when $B = T$ it includes all buyers. The optimal online expected reward is $\sup_\pi \mathbb{E}[R^\pi]$. For a *particular* policy, expected regret means

$$\mathrm{Reg}(\pi) = \mathbb{E}[\mathrm{OPT}(V_1, \dots, V_T) - R^\pi]. \tag{11.1}$$

The clairvoyant, the best online policy, and a proposed online policy are distinct benchmarks.

**Brief static-price review.** For continuous $F$, a static threshold $p_L$ satisfying $\mathbb{P}(V_t \geq p_L) = B/T$ gives $N(p_L) \sim \text{Binomial}(T, B/T)$ qualifying buyers. Its standard deviation is at most $\sqrt{T}/2$. For bounded values, this count error yields the $O(\sqrt{T})$ fluid comparison developed in Lectures 9–10; their two-value instance shows a matching gap for the specified static policy. The board shorthand $\mathbb{E}[\text{OPT}] \leq Bp_L$ is not generally true: hindsight may select buyers with values above $p_L$.

## 11.3 Compensated coupling: forcing hindsight

The Simons talk considers two controllers on one sample path. ONLINE knows current and past observations. OFFLINE knows the entire realized future. Whenever OFFLINE does not wish to take ONLINE's action, pay it the decrease in its optimal continuation value and force it to follow ONLINE.

Fix a complete sequence $\omega = (v_1, \ldots, v_T)$. Let $H_t(b; \omega)$ be the sum of the $\min\{b, T - t + 1\}$ largest values from periods $t, \ldots, T$, with $H_{T+1}(b; \omega) = 0$. Thus excess capacity simply remains unused. For feasible actions define

$$\begin{aligned} Q_t^{\text{off}}(b, 1; \omega) &= v_t + H_{t+1}(b - 1; \omega), \\ Q_t^{\text{off}}(b, 0; \omega) &= H_{t+1}(b; \omega), \qquad H_t(b; \omega) = \max_{a \in \{0,1\}: a \leq b} Q_t^{\text{off}}(b, a; \omega). \end{aligned} \tag{11.2}$$

This is $V^{\text{off}}(t, s)$ and its action-value $Q^{\text{off}}(t, s, a)$ from the talk, with state $s = (t, b)$. Both depend on the *realized* future; neither is the expected online dynamic-programming value.

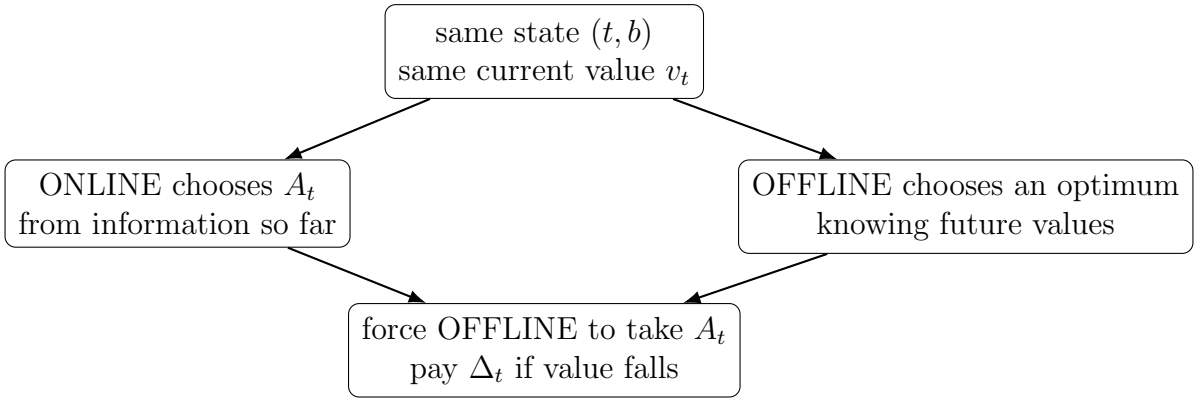



**Figure 11.1.** Compensated coupling evaluates ONLINE's action against the offline optimum at the *same* remaining capacity and realized future.

Let $B_t$ be ONLINE's capacity before period $t$: $B_1 = B$, $B_{t+1} = B_t - A_t$. The smallest necessary compensation is

$$\Delta_t = H_t(B_t; \omega) - Q_t^{\text{off}}(B_t, A_t; \omega) \geq 0. \tag{11.3}$$

The chosen action earns $A_t v_t$ and reaches state $(t+1, B_{t+1})$, so

$$H_t(B_t;\omega) = A_t v_t + H_{t+1}(B_{t+1};\omega) + \Delta_t.$$

Summing over $t$ cancels successive continuation values. Since $H_1(B;\omega) = \mathrm{OPT}(\omega)$ and $H_{T+1} = 0$,

$$\boxed{\mathrm{OPT}(\omega) - R^\pi(\omega) = \sum_{t=1}^{T} \Delta_t(\omega),} \qquad \mathrm{Reg}(\pi) = \sum_{t=1}^{T} \mathbb{E}[\Delta_t]. \tag{11.4}$$

The talk's more general comparison is an inequality for other benchmarks. With this exact hindsight optimum and minimal compensation, it is a *pathwise equality*. No formal information-flow notation is needed to verify the cancellation.

## 11.4 Satisfying actions and ties

An action is *satisfying* if some optimal hindsight allocation from the current state takes it. Equivalently, its $Q_t^{\mathrm{off}}$ value equals $H_t$, so $\Delta_t = 0$. Multiple actions can be satisfying. Predicting one arbitrarily tie-broken hindsight allocation would mistake a costless alternative for an error.

**Example 11.1 (A marginal buyer).** *There are two seats, the current value is* 1*, and future values are* $(2, 1, 0)$*. Hindsight earns* 3 *using either the current or the future value-*1 *buyer: accept and reject are both satisfying. With future* $(2, 2, 0)$*, only reject is satisfying; with future* $(0, 0, 0)$*, only accept is satisfying.*

If values are bounded by $v_{\max}$, then a non-satisfying action costs at most $v_{\max}$:

$$\mathrm{Reg}(\pi) \le v_{\max} \sum_{t=1}^{T} \mathbb{P}(A_t \text{ is not satisfying at ONLINE's state}). \tag{11.5}$$

For continuous values, measuring the *size* of the mismatch gives a sharper calculation.

## 11.5 Uniform values: the October 6 proof

Assume $V_t \overset{\text{iid}}{\sim} \mathrm{Uniform}[0,1]$. At the current period, let $b$ seats remain and let $h = T - t$ be the number of buyers *strictly after* the current one. For $1 \le b \le h$, let $W_{h,b}$ be the $b$-th largest of those $h$ future values. It is random to ONLINE but fixed on a realized future path. Equation (11.2) gives

$$Q_t^{\mathrm{off}}(b,1) - Q_t^{\mathrm{off}}(b,0) = V_t - W_{h,b}.$$

Thus hindsight accepts if $V_t > W_{h,b}$ and rejects if $V_t < W_{h,b}$. If $b = 0$, reject is forced; if $b \geq h + 1$, accept all remaining buyers. These boundary actions pay zero compensation.

Suppose ONLINE accepts when $V_t \geq \theta$, where $\theta$ depends only on $(h, b)$. A disagreement occurs only when $V_t$ lies between the online threshold and the future order statistic. Its exact cost is

$$\Delta_t = |W_{h,b} - V_t| \, \mathbf{1}\{\min(\theta, W_{h,b}) < V_t < \max(\theta, W_{h,b})\}. \tag{11.6}$$

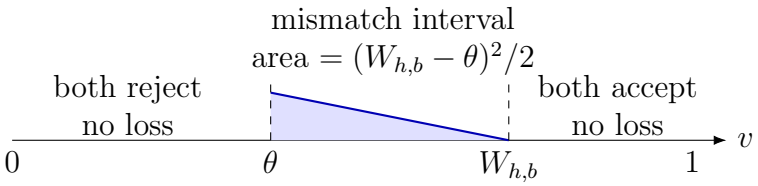



**Figure 11.2.** When $\theta < W_{h,b}$, ONLINE accepts a buyer whom hindsight rejects. The vertical height is the lost value $W_{h,b} - v$. Reversing the threshold order gives the same triangle.

Condition on $W_{h,b} = w$. The current uniform value is independent of the future. If $\theta < w$, integrate its density over the interval:

$$\mathbb{E}[\Delta_t \mid W_{h,b} = w] = \int_\theta^w (w - v)\, dv = \frac{(w - \theta)^2}{2}.$$

If $w < \theta$, the integral is $\int_w^\theta (v - w)\, dv = (\theta - w)^2/2$. Therefore

$$\boxed{\mathbb{E}[\Delta_t \mid W_{h,b}] = \frac{1}{2}(W_{h,b} - \theta)^2.} \tag{11.7}$$

The triangle's *area*, rather than the probability of disagreement alone, is the economic loss.

## 11.6 Order statistics and the logarithmic sum

The $b$-th largest among $h$ uniforms is the $(h - b + 1)$-st smallest, so $W_{h,b} \sim \text{Beta}(h - b + 1, b)$. Its moments are

$$\mathbb{E}[W_{h,b}] = \frac{h - b + 1}{h + 1} = 1 - \frac{b}{h + 1}, \qquad \text{Var}(W_{h,b}) = \frac{b(h - b + 1)}{(h + 1)^2(h + 2)}. \tag{11.8}$$

For $b = 1$ this is the maximum, with mean $h/(h + 1)$; for $b = h$ it is the minimum, with mean $1/(h + 1)$.

For any deterministic $\theta$ chosen from state $(h, b)$,

$$\mathbb{E}[(W_{h,b} - \theta)^2] = \text{Var}(W_{h,b}) + (\mathbb{E}[W_{h,b}] - \theta)^2.$$

Thus the exact minimizer of the one-step *expected compensation* is

$$\boxed{\theta_{h,b} = \mathbb{E}[W_{h,b}] = 1 - \frac{b}{h+1}.} \tag{11.9}$$

This is the uniform quantile at the remaining capacity fraction $b/(h+1)$ among the $h+1$ opportunities *including* the current buyer. The classroom shorthand $1 - b/h$ uses only the $h$ future buyers; for $h \geq 1$ it differs by $b/[h(h+1)] \leq 1/(h+1)$. The exact choice makes the squared-error calculation clean.

Since $b(h-b+1) \leq (h+1)^2/4$, equations (11.7) and (11.8) give

$$\mathbb{E}[\Delta_t \mid B_t = b] = \tfrac{1}{2}\operatorname{Var}(W_{h,b}) \leq \frac{1}{8(h+2)} \qquad (1 \leq b \leq h). \tag{11.10}$$

The future values remain independent of $B_t$, which depends only on the past.

**Theorem 11.2 (Uniform-value threshold policy).** *Suppose values are iid uniform on $[0,1]$, $0 \leq B \leq T$, and the policy rejects if $b = 0$, accepts if $b \geq h+1$, and otherwise accepts exactly when $V_t \geq 1 - b/(h+1)$. Then*

$$\operatorname{Reg}(\pi) \leq \frac{1}{8}\sum_{h=1}^{T-1}\frac{1}{h+2} \leq \frac{1}{8}H_{T+1} = O(\log T), \tag{11.11}$$

*where $H_n = \sum_{k=1}^{n} 1/k$.*

**Proof:** Apply the pathwise identity (11.4), then (11.10) for each remaining horizon $h$. Boundary states have zero compensation. ■

This upper bound also applies to the best online policy. Bray's separate lower bound shows logarithmic regret is the right order for uniform values when capacity is a fixed nontrivial fraction of the horizon. This is a continuous-value result, not an $O(1)$ finite-type theorem.

**Remark 11.3 (Mean versus median).** *The rule (11.9) minimizes expected* compensation. *An ideal selector that minimizes only the* probability *of an unsatisfying action would compare $V_t$ with a median of $W_{h,b}$. These thresholds need not coincide. The mean rule is the model-based quantile policy analyzed above.*

For general continuous $F$, the analogous capacity-fraction quantile is $F^{-1}(1 - b/(h+1))$ when defined. The future order statistic has distribution $F^{-1}(U_{(h-b+1)})$, whose mean need not equal this quantile. The uniform proof does not automatically establish $O(\log T)$ for every continuous $F$.

# 11.7 The Bayes Selector with discrete values: October 8

Now let $V_t$ take one of $K$ *positive* values $v_1 > v_2 > \cdots > v_K > 0$, with iid probabilities $p_i = \mathbb{P}(V_t = v_i) > 0$. The online policy again has remaining capacity $B_t$ just before seeing buyer $t$. Write $b = B_t$, let $h = T - t$ count buyers *strictly after* the current one, and let $m = h + 1 = T - t + 1$ count all opportunities *including* this buyer. Put $\bar{p}_i = \sum_{j=1}^{i} p_j$ and $\bar{p}_0 = 0$.

The online decision is still accept or reject. Hindsight accepts the highest-valued buyers up to capacity $b$. We want an online decision that is unlikely to disagree with *every* optimal hindsight continuation, even though future type counts are unknown.

## A re-solved fluid prediction

At each state, replace the unknown remaining type counts by their unconditional means $mp_i$ and solve

$$\begin{aligned} \max_{y \in \mathbb{R}^K} \quad & \sum_{i=1}^{K} v_i y_i \\ \text{subject to} \quad & \sum_{i=1}^{K} y_i \le b, \\ & 0 \le y_i \le mp_i \qquad (i = 1, \ldots, K). \end{aligned} \tag{11.12}$$

Here $y_i$ predicts how many type-$i$ buyers hindsight would take. The LP uses the $m$ opportunities including the current one, as in the October 8 board. After conditioning on the current buyer's type, the *future* count instead has mean $hp_i$; we will use that distinction in the probability calculation. Re-solving with the *current* $b$ is what makes this an adaptive rather than a static fluid policy.

Since $v_1 > \cdots > v_K$, an exchange argument fills higher-value types first. Its solution is

$$y_i^* = \min\left\{mp_i,\, [b - m\bar{p}_{i-1}]^+\right\}. \tag{11.13}$$

Thus a type is fully selected ($y_i^* = mp_i$), excluded ($y_i^* = 0$), or the one marginal type with $0 < y_i^* < mp_i$. The geometry is a cutoff in cumulative probability mass.

## The classroom half-threshold rule

If $b = 0$, reject. If $b > 0$, *always accept type 1*: no future buyer has a strictly greater value, so this action is always satisfying for hindsight. For an arriving type $i > 1$, accept when at least

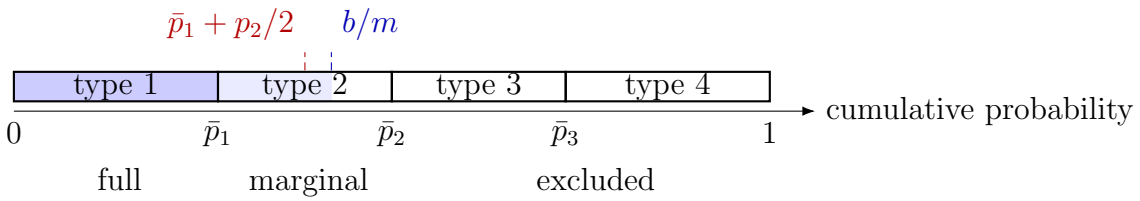



**Figure 11.3.** The fluid LP fills types from highest to lowest value. In this schematic, type 2 is marginal and $b/m$ lies beyond the halfway point of its probability interval, so the half-threshold rule accepts type 2. Positions are illustrative.

half of its expected fluid demand is selected:

$$\frac{y_i^*}{mp_i} \geq \frac{1}{2} \quad \Longleftrightarrow \quad \frac{b}{m} \geq \bar{p}_{i-1} + \frac{p_i}{2}; \qquad \text{otherwise reject.} \tag{11.14}$$

At equality, we accept. A literal half-threshold applied to type 1 would sometimes reject the very best type; the always-accept clause is an essential part of the policy proved below. The fluid fraction is a tractable prediction, *not* an exact conditional probability of hindsight acceptance.

For two types, $v_1 > v_2$,

$$y_1^* = \min\{mp_1, b\}, \qquad y_2^* = \min\{mp_2, [b - mp_1]^+\}.$$

Type 1 is accepted whenever capacity remains. Type 2 is accepted precisely when $b/m \geq p_1 + p_2/2$. If capacity is large relative to expected high-value demand, selling to type 2 is sensible; if capacity is scarce, the rule protects it for type 1. This is the same economic question as Littlewood's rule, now revisited after every arrival.

## What makes an action non-satisfying?

Condition on $B_t = b > 0$ and $V_t = v_i$. Define counts among the $h$ strictly future buyers:

$$N^+_{<i}(t) = \sum_{s=t+1}^{T} \mathbf{1}\{V_s > v_i\}, \qquad N^+_{\leq i}(t) = \sum_{s=t+1}^{T} \mathbf{1}\{V_s \geq v_i\}. \tag{11.15}$$

The subscripts refer to the *value ranking*: $N^+_{<i}$ counts strictly better types $1, \ldots, i-1$; $N^+_{\leq i}$ also includes type $i$. Their distributions are Binomial$(h, \bar{p}_{i-1})$ and Binomial$(h, \bar{p}_i)$, respectively.

Accepting the current buyer is satisfying if $N^+_{<i}(t) \leq b - 1$: after taking all strictly better buyers, there is a seat that can go to the current one. Rejecting is satisfying if $N^+_{\leq i}(t) \geq b$: future buyers at least as valuable can fill all seats. Consequently, the *exact* bad events are

$$\begin{aligned} &\text{accept is non-satisfying} \quad \Longleftrightarrow N^+_{<i}(t) \geq b, \\ &\text{reject is non-satisfying} \quad \Longleftrightarrow N^+_{\leq i}(t) \leq b - 1. \end{aligned} \tag{11.16}$$

When both actions are satisfying, the current buyer is marginal: one optimal continuation accepts this buyer and another takes an equal-valued future buyer instead. We need not predict which tie-breaking hindsight allocation will be chosen. If $b = 0$, rejection is forced and has zero compensation.

## A wrong action requires a large count deviation

Fix $i > 1$, $b > 0$, and $h \geq 1$; recall $m = h + 1$. Future types are independent of the current state and arrival type. If the rule *accepts*, then $b \geq m(\bar{p}_{i-1} + p_i/2)$. The bad-accept event in (11.16) requires the higher-type count to exceed its mean $h\bar{p}_{i-1}$ by at least

$$b - h\bar{p}_{i-1} \geq \frac{mp_i}{2} + \bar{p}_{i-1} \geq \frac{mp_i}{2}.$$

If the rule *rejects*, then $b < m(\bar{p}_i - p_i/2)$. The bad-reject event requires the count of types $i$ or better to fall below its mean $h\bar{p}_i$ by more than

$$h\bar{p}_i - (b-1) > \frac{mp_i}{2} + 1 - \bar{p}_i \geq \frac{mp_i}{2}.$$

Hoeffding's binomial bound, in the same clean-event spirit as the UCB proof, now gives *for either action*

$$\mathbb{P}(\text{non-satisfying action} \mid B_t = b, V_t = v_i) \leq \exp\left(-\frac{m^2 p_i^2}{2h}\right) \leq e^{-mp_i^2/2}. \tag{11.17}$$

For $h = 0$ the policy accepts any positive type if $b > 0$, which is satisfying; for type 1, accepting is always satisfying. The $+1$ in the bad-reject comparison and the distinction between $m$ and $h$ keep the integer and conditioning conventions exact.

## Summing the rare mistakes

One wrong action changes hindsight value by at most $v_{\max} = v_1$, so (11.5) and (11.17) imply a *uniform classroom bound*. Write $p_{\min} = \min_i p_i$. At any period with $m$ opportunities including the current one,

$$\mathbb{P}(\text{non-satisfying action}) \leq \sum_{i=2}^{K} p_i e^{-mp_i^2/2} \leq e^{-mp_{\min}^2/2}.$$

Since a decreasing exponential sum is no larger than its integral starting at zero,

$$\boxed{\mathrm{Reg}(\pi_{\text{fluidBS}}) \leq v_{\max} \sum_{m=1}^{T} e^{-mp_{\min}^2/2} \leq \frac{2v_{\max}}{p_{\min}^2}.} \tag{11.18}$$

This is $O(1)$ in the horizon for a *fixed* finite type distribution. It is not a guarantee uniform over distributions whose smallest type probability shrinks with $T$.

Keeping each factor $p_i$ outside its own geometric series gives the sharper published statement:

**Theorem 11.4 (Finite-type Fluid Bayes Selector; Vera–Banerjee).** *Assume one resource and iid arrivals from $K$ fixed positive value types, each with $p_i > 0$. The re-solved policy (11.14), including its highest-type exception, has expected regret against hindsight bounded by*

$$\mathrm{Reg}(\pi_{\mathrm{fluidBS}}) \le 2v_{\max} \sum_{i=2}^{K} \frac{1}{p_i} \le \frac{2(K-1)v_{\max}}{p_{\min}}. \tag{11.19}$$

Indeed, $\sum_{m\ge 1} e^{-mp_i^2/2} \le 2/p_i^2$. The classroom bound (11.18) uses the single worst-case $p_{\min}$; Theorem 11.4 retains the probability of seeing each type. The ideal Bayes Selector instead minimizes the *exact* conditional probability that its action is non-satisfying. The LP-based Fluid Bayes Selector minimizes tractable *upper bounds* on those probabilities and is the policy analyzed here.

### From Bayes selection back to demand

The principle is to predict an action that hindsight would find satisfying and charge compensation only on rare prediction errors. The October 8 board ended by returning to Littlewood's two-fare example. Littlewood takes high- and low-fare demand as inputs. But a customer who prefers the low fare may buy the high fare when the low fare is unavailable. Which fare is purchased therefore depends on the offered set: this is *substitution*.

What happens when a firm estimates demand from observed sales while its own booking controls determine which fares are available? That question opens Lecture 15 on the Spiral-Down Effect.

## 11.8 Takeaways

1. Compensation is the reduction in hindsight value from forcing an online action; the reductions telescope pathwise.
2. A satisfying action pays zero compensation, even if a different hindsight-optimal action also exists.
3. For uniform values, conditional expected compensation is the triangle area $(W_{h,b} - \theta)^2/2$.
4. The exact mean threshold $1 - b/(h+1)$ has $O(1/h)$ one-step loss and $O(\log T)$ cumulative regret.

5. For fixed finite types, the re-solved LP and half-threshold rule make non-satisfying actions exponentially rare. Their compensation sums to $O(1)$ under Theorem 11.4. The $O(\sqrt{T})$ static, $O(\log T)$ uniform, and $O(1)$ finite-type statements concern their specified policies and assumptions.

6. The October 8 transition asks how booking controls change which fares customers purchase. Lecture 15 develops the resulting Spiral-Down Effect.

# Recommended reading

**Lecture sources.**
Banerjee, *The Compensated Coupling*, IFML–Simons Institute talk (October 2022), slides 14–20 and 24–34; September 29 Zoom notes and the October 6 board.

**Finite types.**
Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, *Management Science* 67 (2021), Sections 3–4, Theorem 2 and Appendix B.3 (preprint); the October 8 board develops the single-resource LP proof.

**Continuous values*.**
Bray, *Logarithmic Regret in Multisecretary and Online Linear Programs with Continuous Valuations*, *Operations Research* (2024), Section 3.